

Problem Set 7: Ridge Regression and Entropy (April 19 - May 2, 18:00)

Problem 1 (Ridge Regression)

Assume given a dataset $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \subseteq \mathbb{R}^d \times \mathbb{R}$. Let $X \in \mathbb{R}^{n \times d}$ be the matrix where row i , for $i \leq n$, is $x_i^\top$. Let $y \in \mathbb{R}^n$ be the vector where entry i , for $i \leq n$, is y_i .

The *ridge estimator* $f(x) = x^\top \tilde{w}_R$ is a linear estimator that can be shown to be more robust to noise in the data and, therefore, less prone to overfitting, when compared to the OLSE. We have the following closed form solution for $\tilde{w}_R$

$$\tilde{w}_R = (X^\top X + \lambda I)^{-1} X^\top y \quad (1)$$

which is derived from the following loss function:

$$L(w) = \lambda \|w\|^2 + \sum_{i \leq n} (y_i - x_i^\top w)^2. \quad (2)$$

where we have the regularization term on top of OLSE. Please note that this is equivalent to OLSE when $\lambda = 0$

Demonstrate that

$$\tilde{w}_R = \arg \min_{w \in \mathbb{R}^d} L(w). \quad (3)$$

Problem 2 (Theory of Information: Entropy)

Consider a discrete random variable X with a set of possible outcomes $\mathcal{X}$, and probabilities $p(x)$ for each outcome $x \in \mathcal{X}$.

The entropy of X is defined as $H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x)$. Note that in the lecture, we defined the entropy for datasets. For a dataset D , the entropy of D is $H(D) = -\sum_{y \leq K} p_y \log p_y$, where p_y is the fraction of examples in the dataset with label $y \leq K$. This definition of entropy for datasets comes from treating D as a random variable X_D with range in $\{1, \dots, K\}$, where $P(X = i) = p_i$.

1) Non-Negativity of Entropy

Show that the entropy $H(X) \geq 0$, for any random variable X .

2) Conditional Entropy

Let now X, Y be a random variable with range in $\mathcal{X} \times \mathcal{Y}$. The conditional entropy of (X, Y) is :

$$H(Y|X) = \sum_{x \in \mathcal{X}} p(x) H(Y|X = x)$$

Show that

$$H(Y|X) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x).$$

3) Chain Rule

Show the following:

$$H(X, Y) = H(X) + H(Y|X).$$

4) Maximization of Entropy: Uniform Distribution

Suppose that $\mathcal{X} = \{x_1, \dots, x_n\}$. Show that $H(X)$ is maximized when X is uniformly distributed, that is, when $p(x_i) = \frac{1}{n}$, for any $i \leq n$.

Hint: You may use without proof the following inequality:

$$-\frac{1}{n} \log \frac{1}{n} \geq -\sum_{i \leq n} \frac{1}{n} p(x_i) \log p(x_i), \quad \text{for any distribution } p.$$

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Demonstrate that

$$\tilde{w}_R = \arg \min_{w \in \mathbb{R}^d} L(w). \quad (3)$$

Remember diff. rules for vectors & matrices
↳ differentiate & set = 0

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Show that

$$H(Y|X) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x).$$

$$0 \leq p(x) \leq 1$$

↳ effect on $\log(p(x))$

$$p(x) \cdot p(y|x) = p(x, y)$$

3) Chain Rule

Show the following:

$$H(X, Y) = H(X) + H(Y|X).$$

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Suppose that $\mathcal{X} = \{x_1, \dots, x_n\}$. Show that $H(X)$ is maximized when X is uniformly distributed, that is, when $p(x_i) = \frac{1}{n}$, for any $i \leq n$.

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$$H_{\text{uniform}}(X) = -\sum_{i=1}^n \frac{1}{n} \log\left(\frac{1}{n}\right)$$